

Status

Boba — Feature Status (all components)

Revision: 10 **Last modified:** 2026-08-21T00:00:00Z **Scope:** Every system component, service, infrastructure piece, and client app of the Boba project — one row per REAL unit (endpoint / handler / client method / component control / plugin / subcommand / script), grouped by component. **Authority:** assembled by READ-ONLY repo inventory (codegraph + grep + source reading) on 2026-06-15, expanded to per-unit granularity (§11.4.118 discovery-pressure) on 2026-06-16. Cross-references AGENTS.md, CLAUDE.md, docs/REMAINING_WORK_PLAN.md.

Rev 9 (2026-08-20, BOB-083 / GOVERNANCE_AUDIT_2026-08-08_ROUND2.md RD2-16) — staleness catch-up (§11.4.44/§11.4.45/ §11.4.86): 161 commits landed between the Rev 8 pass (e6162f7, 2026-08-18) and this pass's HEAD (e0d60ab, 2026-08-20). Real git log e6162f7..HEAD evidence: the overwhelming majority are the BOB-126 forced-logout killpg(-1, SIGKILL) incident chain (root cause found + closed, new universal §11.4.263 anchor) plus surrounding test/gate/quality hardening (BOB-127..135) — process-safety and CI-discipline work, **not new product features**, so no new feature rows are warranted from that set (§11.4.6: verified by reading the commit log, not assumed). Two exceptions **were** feature-relevant and are corrected below with cited live evidence: **RW-06** (RuTracker ReDoS fix deployment) — CLOSED live 2026-08-19 (1c0389a, BOB-093); **RW-07** (nnmclub /auth/nnmclub/status SOURCE→ARTIFACT drift) — CLOSED live 2026-08-19 (7baef2b, BOB-092). Both rows below updated to PASS – LIVE with their real evidence paths. docs/REMAINING_WORK_PLAN.md itself was NOT edited (out of this task's owned-file scope — see the Status_Summary companion for the plain-language version of this note). Total features cataloged: **unchanged at 292** (no rows added or removed this pass, only 2 existing rows' Validation cells corrected). Every other pre-existing row was left untouched

— a full per-unit re-audit against the 161-commit delta was judged out of the bounded scope of this staleness-catch-up pass, per the same §11.4.6 option-B discipline Rev 8 used.

Rev 8 (2026-08-18, BOB-075) — staleness remediation (§11.4.44 / §11.4.86 /

GOVERNANCE_AUDIT_2026-08-08_ROUND2.md

RD2-08): the audit’s staleness claim was CONFIRMED, not a false positive — a real content edit landed in ea86ce1 (2026-08-15, the docs_chain-engine row swap in §8d below) WITHOUT the Revision/Last-modified header being bumped, and 60 commits landed between the Rev 7 content commit (646b295, 2026-06-16) and this pass (64705eb, 2026-08-18) that were never reflected at row-granularity. This pass performed a TARGETED refresh (§11.4.6 option B — this document is narrative, “assembled by READ-ONLY repo inventory,” not mechanically derivable from a single source-of-truth table, so a full per-unit re-audit of all 289 pre-existing rows was judged out of the bounded scope of a staleness-fix task and is NOT claimed here): added 3 new rows to §8d for scripts that shipped since Rev 7 and were completely absent from this ledger — scripts/load-tracker-cookies.sh (tracker-cookies autoloader, 619a5f6), scripts/boba-svc.sh (sudo-free systemd wrapper, 444d94e), scripts/commit-push-all.sh (§11.4.234 dedicated commit/push entrypoint, 08e870e) — each cited to its real source file + covering challenge. Total features cataloged: 289 → **292**. Every pre-existing row was left untouched (no content was invented to “look fresh”); the honest scope boundary of this pass is recorded under “Components NOT inventoried” below.

Rev 7 (2026-06-16) — live-PASS update for the search/auth fix batch (HEAD 7e9cab5): the multi-word URL-

encoding fix (da7d709, ~17 nova3 plugins), the new RUTRACKER_COOKIES injection (2fc29fc), NNMCLUB_COOKIES auth, /auth/status cookie reflection (9c2f8dc), the new GET /api/v1/healthz JSON endpoint (137d7ff), and §11.4.85 stress+chaos coverage for the multi-word fix (7e9cab5) were all PROVEN end-to-end on the live nezha stack — full-fleet the matrix = **2600 results / 23 contributing trackers / zero plugin_bad_query_encoding**, all four private trackers authenticate (rutracker 50, nnmclub 50, kinozal 50, iptorrents 49).

Evidence: docs/qa/search-fix-verify-20260616/. kickass.py reclassified **Won’t-fix structurally-impossible**

(§11.4.112) per docs/research/kickass_403_20260616/. NOTE (§11.4.6): no Go files were touched this session — the Go rows are unchanged (the pre-existing magnet_stress_chaos_test.go guard stands; no NEW Go query-encoding guard was added this session, so none is claimed here).

Captured-evidence-driven (§11.4.5 / §11.4.45 / §11.4.86). Every feature row cites a real source file (file:line where load-bearing), endpoint, command, or control. **No invented features** (§11.4.6). As of Rev 5 the **Video** column is DEFINITIVE for every row — there is no bare PENDING. Each cell is exactly one of: VIDEO-CONFIRMED – <file> (the feature is shown on-screen in a committed recording), PENDING (UI – film next) (a user-visible control/dialog not yet *individually* filmed), or N/A (no UI – test-covered + exercised by <journey>) (a non-user-visible unit — REST endpoint / handler / client method / parser / scanner / service / crypto / repo / script — which has no screen of its own and is confirmed by its cited tests plus the UI/CLI journey that drives it). **Honest classification (§11.4.6/§11.4.143): a row is VIDEO-CONFIRMED only when a real recording actually shows it.**

Per-row Video tally (291 feature rows — the pre-existing Rev ≤7 tally of 288 vs the stated “289” total was a 1-row pre-existing discrepancy not reconciled by this pass, §11.4.6 honest note; +3 for the Rev 8 §8d additions, all N/A/host-script): **28 VIDEO-CONFIRMED, 14 PENDING (UI — film next), 249 N/A (no UI — test-covered).**

Column legend: - **Impl** = implemented / partial / stub / missing - **Wired** = reachable on a running default-profile stack (yes / no / opt-in / host) - **Tests** = test types present + cite file (unit / integration / e2e / security / stress-chaos / property / contract); none if no covering test found - **Validation** = honest validation posture (tested-green-in-suite / unit-only / live-evidence-captured / not-validated / operator-blocked) - **Video** = DEFINITIVE video-recording status — one of VIDEO-CONFIRMED – <file> / PENDING (UI – film next) / N/A (no UI – test-covered + exercised by <journey>)

Video-recording confirmations (§11.4.143 / §11.4.83)

Real-use recordings (Claude-vision analyzed — the HelixAgent ensemble vision is a proven stub, see docs/qa/recording-readiness-20260615/) in /Volumes/T7/Downloads/Recordings/ (project-prefix boba-). Per-row Video cells stay PENDING where a feature has not been *individually* filmed; the recordings below confirm the headline user-facing flows on-screen.

Recording	Surface	Features confirmed	Verdict
boba-cli-orchestrator-demo.mp4			

Recording	Surface	Features confirmed	Verdict
boba-web-search-flow.mp4	CLI (boba-ctl)	status / health / list — 4 services running & healthy	docs/qa/recordings-20260615/boba-cli-verdict.md
	Web dashboard	launch → search “debian” → live results → “829 results (288 merged)” complete dashboard load (29 trackers, qBit connected), result rows render qBit + Download buttons (clickable), tab nav (Results/ Active Downloads/ Trackers/ Schedules/ Hooks), Jackett / jackett credentials page, theme content-script	boba-web-search-flow-verdict.md
boba-web-feature-tour.mp4 + boba-web-dashboard-tour.mp4	Web dashboard	(clickable), tab nav (Results/ Active Downloads/ Trackers/ Schedules/ Hooks), Jackett / jackett credentials page, theme content-script	boba-web-feature-tour-verdict.md
boba-extension-{scan-detect,popup,options,tour}.mp4	BobaLink extension (MV3, real --load-extension Chromium)	detects a magnet + appends a MAGNET badge (non-torrent correctly skipped);	boba-extension-verdict.md

Recording	Surface	Features confirmed	Verdict
boba-web-tabs-theme-tour.mp4 (+ frames boba-web-tab-active-downloads.png / boba-web-tab-schedules.png / boba-web-tab-hooks.png / boba-web-theme-toggle.png) + boba-web-jackett-add-credential.png	Web dashboard + Jackett page	popup UI (header, server status, Refresh + Send All, Detected-torrents); options page (7 settings tabs + server config) Active Downloads / Schedules / Hooks tabs each render their empty-state + columns; theme toggle; processing spinner ("Found N results..."); bridge/qBit-Connected header indicator; Jackett Add-credential dialog (Name/ Username/ Password/ Cookies + Cancel/ Save)	Claude-vision verified 2026-06-16

Honest scope (§11.4.6, no bluff): - Backend services (download-proxy / merge / boba-jackett / Go) verified **200 server-side inside the VM** and exercised through the web UI recordings. - **Console errors** seen in the web recordings are an **SSH-tunnel SSE/poll artifact** (macOS↔VM), NOT product

defects — every endpoint is 200 in the VM. See the feature-tour verdict. - **BobaLink extension: VIDEO-CONFIRMED** — the prior --load-extension tooling gap was closed by building extension/scripts/record-features.mjs (loads the built MV3 artifact into a real Chromium, records the scan/popup/options journeys). Plus its passing test suite. - **TUI / mobile / desktop:** do not exist in this project (server-side + Angular web + browser extension only) — nothing to record.

Component summary

#	Component	Service / Port	Features cataloged
1	Download Proxy + Merge Search Service (Python/FastAPI)	qbittorrent-proxy :7186 / :7187	69
2	qBitTorrent-go backend (Go/Gin)	qbittorrent-proxy-go :7186/:7187/:7188 (opt-in --profile go)	47
3	boba-jackett (Go)	boba-jackett :7189	26
4	Tracker plugins (plugins/*.py)	run inside qbittorrent-proxy	30
5	Angular 21 frontend dashboard (frontend/)	served from :7187	34
6	BobaLink browser extension (extension/)	WXT MV3 client	39
7	WebUI bridge (webui-bridge.py)	host process :7188	4
8	Infrastructure / CLI / shell scripts	host	43

Total features cataloged: 292 (Rev 8: +3 — scripts/load-tracker-cookies.sh, scripts/boba-svc.sh, scripts/commit-push-all.sh; Rev 7: +1 — new GET /api/v1/healthz endpoint)

1. Download Proxy + Merge Search Service (Python/FastAPI) — :7186 / :7187

Entry: download-proxy/src/main.py (starts legacy proxy thread + uvicorn api:app). This is the **shipped default product**.

1a. Merge Search Service — REST API (download-proxy/src/api/routes.py)

Feature	Impl	Wired	Tests	Valid
POST /api/v1/search — async fan-out search, returns search_id (routes.py:283)	implemented	yes	integration tests/integration/test_merge_api.py; e2e test_full_pipeline.py; stress test_merge_search_stress_chaos.py	tested green suite LIVE
POST /api/v1/search/sync — synchronous search (routes.py:377)	implemented	yes	integration test_merge_api.py; stress test_search_stress.py	tested green suite reset tunnel (RW-
GET /api/v1/search/stream/{id} — SSE result stream (routes.py:547)	implemented	yes	integration test_realtime_streaming.py, test_streaming_browser.py	tested green suite
GET /api/v1/search/{id} — poll search state (routes.py:586)	implemented	yes	integration test_merge_api.py	tested green suite
POST /api/v1/search/{id}/abort — cancel in-flight search (routes.py:619)	implemented	yes	integration test_merge_api.py	tested green suite
GET /api/v1/downloads/active — qBit active torrents (routes.py:629)	implemented	yes	integration test_buttons_api.py	tested green suite

Feature	Impl	Wired	Tests	Valid
POST /api/v1/auth/qbittorrent — qBit WebUI login (routes.py:671)	implemented	yes	integration test_login_actions.py	tested green suite
POST /api/v1/download — add torrent to qBit (magnet/url) (routes.py:958)	implemented	yes	integration test_button_functions.py, test_buttons_api.py; security test_ssrf_and_magnet_auth.py	tested green suite SSRF RW-C
POST /api/v1/download/upload — upload .torrent file to qBit (routes.py:1123)	implemented	yes	unit tests/unit/api_layer/test_download_upload.py	unit-integ
POST /api/v1/download/file — fetch tracker .torrent w/ auth cookies (routes.py:1212)	implemented	yes	integration test_buttons_api.py; security test_ssrf_and_magnet_auth.py	tested green suite SSRF RW-C
POST /api/v1/magnet — resolve result → magnet link (routes.py:1279)	implemented	yes	integration test_magnet_dialog.py, test_buttons_api.py	tested green suite gap I
GET /api/v1/theme — read theme state (routes.py:61)	implemented	yes	contract test_crossapp_theme_contract.py	tested green suite
PUT /api/v1/theme — persist theme (routes.py:67)	implemented	yes	e2e test_theme_runtime.py, test_crossapp_theme.py	tested green suite
GET /api/v1/theme/stream — SSE theme broadcast (routes.py:76)	implemented	yes	e2e test_crossapp_theme.py	tested green suite

1b. Auth API — per-tracker (download-proxy/src/api/auth.py)

Feature	Impl	Wired	Tests	Validation
GET /auth/ rutracker/ status — RuTracker session state (auth.py:56)	implemented	yes	integration test_auth_and_links.py; e2e test_live_stack_evidence.py	partial — (BOB-008)
GET /auth/ rutracker/ captcha — fetch CAPTCHA image (auth.py:108)	implemented	yes	integration test_auth_and_links.py	partial —
POST /auth/ rutracker/ login — username/ password + CAPTCHA login (auth.py:214)	implemented	yes	integration test_auth_and_links.py	partial —
POST /auth/ rutracker/ cookie-login — cookie- based login (auth.py:298)	implemented	yes	integration test_auth_and_links.py	PASS — I RUTRACKER_ authenticat matrix = 5 (docs/qa/s §1)
GET /auth/ nnmclub/ status — NNM-Club session state (auth.py:350)	implemented	yes	e2e test_live_stack_evidence.py:265 (no longer skips — see Rev 9)	PASS — I (BOB-09) SOURCE- live probe HTTP 200 JSON (doc nnmclub_st the e2e's removed (FAILs on r silently sk
POST /auth/ nnmclub/ login — NNM-Club	implemented	yes	e2e test_live_stack_evidence.py	partial —

Feature	Impl	Wired	Tests	Validation
login (auth.py:407)				
GET /auth/ status — all- tracker aggregate session state (auth.py:479)	implemented	yes	integration test_auth_state_ui.py; auth- status cookie-reflection test (\$1.1)	PASS — I reflects RU NNMCLUB_CO (9c2f8dc), rutracker/ search-fix
POST /auth/ qbittorrent/ logout — qBit WebUI logout (auth.py:532)	implemented	yes	integration test_login_actions.py	tested-gre
Kinozal auth (status/ login) REST route	missing	no	none — no kinozal/login route in auth.py	not-imple cookies vi
IPTorrents auth (status/ login) REST route	missing	no	none — no iptorrents/login route in auth.py	not-imple flow via Ja

1c. Search orchestration internals (download-proxy/ src/merge_service/search.py)

Feature	Impl	Wired	Tests
Concurrency caps MAX_CONCURRENT_SEARCHES (429 on saturation)	implemented	yes	concurrency test_orchestrator_semaphore
Per-search tracker semaphore MAX_CONCURRENT_TRACKERS	implemented	yes	concurrency test_tracker

Feature	Impl	Wired	Tests
Public-tracker subprocess fan-out (NDJSON via patched novaprinter)	implemented	yes	e2e test_public_trackers_ret stress test_search_orchestration
Public-tracker deadline kill PUBLIC_TRACKER_DEADLINE_SECONDS	implemented	yes	chaos test_tracker_failu
Private-tracker aiohttp+cookie path (rutracker/kinozal/nnmclub/iptorrents)	implemented	yes	integration test_tracker_ (live-gated)
Per-tracker TrackerSearchStat real-time stats	implemented	yes	contract test_tracker_sta property test_tracker_stats_prope
Stderr error classification (_classify_plugin_stderr)	implemented	yes	chaos test_tracker_failu
	implemented	yes	

Feature	Impl	Wired	Tests
TTL-bounded data stores (cachetools) — no leak			memory test_orchestrator_caches_

1d. Deduplication (download-proxy/src/merge_service/deduplicator.py)

Feature	Impl	Wired	Tests	Validation
Tier 1 dedup — metadata (imdb/tmdb) key	implemented	yes	property test_deduplicator_properties.py; benchmark test_deduplication_benchmark.py	tested-green-in-suite
Tier 2 dedup — infohash key	implemented	yes	property test_deduplicator_properties.py	tested-green-in-suite
Tier 3 dedup — name+size key	implemented	yes	property test_deduplicator_properties.py	tested-green-in-suite

Feature	Impl	Wired	Tests	Validation
Tier 4 dedup — fuzzy Levenshtein name match	implemented	yes	property test_deduplicator_properties.py; benchmark test_deduplication_benchmark.py	tested-green-in-suite

1e. Enrichment — per metadata source (download-proxy/src/merge_service/enricher.py)

Feature	Impl	Wired	Tests	Validation
Enrichment: OMDb (enricher.py:113)	implemented	yes	stress test_enricher_stress_chaos.py	tested-green-in-suite; needs OMDb_API
Enrichment: TMDB	implemented	yes	stress test_enricher_stress_chaos.py	tested-green; needs TMDB_API
Enrichment: TVMaze	implemented	yes	stress test_enricher_stress_chaos.py	tested-green-in-suite

Feature	Impl	Wired	Tests	Validation
Enrichment: AniList	implemented	yes	stress test_enricher_stress_chaos.py	tested- green-in- suite
Enrichment: MusicBrainz	implemented	yes	stress test_enricher_stress_chaos.py	tested- green-in- suite
Enrichment: OpenLibrary	implemented	yes	stress test_enricher_stress_chaos.py	tested- green-in- suite

1f. Tracker validation (download-proxy/src/merge_service/validator.py)

Feature	Impl	Wired	Tests	Validation	Video
Tracker validation HTTP scrape BEP 48 (validator.py:101)	implemented	yes	none dedicated found (covered via merge tests)	not-validated (no dedicated test)	N/A (no UI — exercised by web search flow seeds/leechers column)
Tracker validation UDP scrape BEP 15 (validator.py:162)	implemented	yes	none dedicated found	not-validated (no dedicated test)	N/A (no UI — exercised by web search flow seeds/leechers column)

1g. Hooks (download-proxy/src/api/hooks.py + merge_service/hooks.py)

Feature	Impl	Wired	Tests
GET /api/v1/hooks — list hooks (hooks.py:105)	implemented	yes	unit test_hooks_coverage concurrency test_hooks_race.py; SE test_hooks_schedules_
POST /api/v1/hooks — create hook (hooks.py:111)	implemented	yes	unit test_hooks_coverage concurrency test_hooks_race.py
DELETE /api/v1/hooks/{id} — delete hook (hooks.py:167)	implemented	yes	unit test_hooks_coverage

Feature	Impl	Wired	Tests
GET /api/v1/hooks/logs — hook execution logs (hooks.py:179)	implemented	yes	unit test_hooks_covera
Hook event dispatch (8 events: search_start..validation_complete) (hooks.py:32)	implemented	yes	concurrency test_hooks_race.py
Hook subprocess execution + sandbox (merge_service/hooks.py)	implemented	yes	security test_no_shell_injecti

1h. Scheduler (download-proxy/src/api/scheduler.py + merge_service/scheduler.py)

Feature	Impl	Wired	Tests	Validat
GET /api/v1/schedules — list schedules (scheduler.py:39)	implemented	yes	security test_hooks_schedules_auth.py	tested-green-in suite; au gap RW
POST /api/v1/schedules — create schedule (scheduler.py:64)	implemented	yes	security test_hooks_schedules_auth.py	tested-green-in suite; au gap RW
GET /api/v1/schedules/{id} — get schedule (scheduler.py:87)	implemented	yes	security test_hooks_schedules_auth.py	tested-green-in suite

Feature	Impl	Wired	Tests	Validation
PATCH /api/v1/schedules/{id} — update schedule (scheduler.py:108)	implemented	yes	security test_hooks_schedules_auth.py	tested-green-in-suite
DELETE /api/v1/schedules/{id} — delete schedule (scheduler.py:126)	implemented	yes	security test_hooks_schedules_auth.py	tested-green-in-suite
Scheduler engine (load/save/tick driver) (merge_service/scheduler.py)	implemented	yes	integration test_dashboard_automation.py	tested-green-in-suite

1i. App-level routes / health / config / dashboard (download-proxy/src/api/__init__.py)

Feature	Impl	Wired	Tests	Validation
GET /api/v1/healthz — machine-parsable JSON health probe (routes.py:37)	implemented	yes	integration test_merge_api.py	PASS — LIVE 2026-06 — NEW endpoint (137d7ff): bare /healthz w swallowed by the SP catch-all, now mounted under /api/v1 return JSON (\$11.4.43)
GET /health (__init__.py:168)	implemented	yes	integration test_live_containers.py	tested-green-in-suite

Feature	Impl	Wired	Tests	Validation
GET /api/v1/bridge/health (__init__.py:173)	implemented	yes	integration test_bridge_root_liveness.py	tested-green-in-suite
GET /api/v1/config (__init__.py:247)	implemented	yes	integration test_merge_api.py	tested-green-in-suite
GET /api/v1/stats (__init__.py:276)	implemented	yes	observability test_metrics_exist.py	tested-green-in-suite
Jinja2 dashboard GET / (__init__.py:337)	implemented	yes	integration test_dashboard_rendering.py	tested-green-in-suite
Jinja2 dashboard GET /dashboard (__init__.py:342)	implemented	yes	integration test_dashboard_rendering.py	tested-green-in-suite
SPA catch-all GET /{path:path} (__init__.py:347)	implemented	yes	integration test_dashboard_rendering.py	tested-green-in-suite

Feature	Impl	Wired	Tests	Validation

1j. Supporting subsystems (Python)

Feature	Impl	Wired	Tests	Validation
Retry helpers (merge_service/ retry.py)	implemented	yes	covered indirectly	
Jackett autoconfig (Python, merge_service/ jackett_autoconfig.py) — legacy; canonical is boba-jackett	implemented	yes	contract test_jackett_autoconfig_contract.py e2e test_jackett_autoconfig_e2e.py benchmark test_jackett_autoconfig_perf.py	
Credential scrubbing log filter (config/ log_filter.py)	implemented	yes	security test_credential_scrubbing.py	

1k. Legacy download proxy (plugins/download_proxy.py:7186)

Feature	Impl	Wired	Tests	Validation
HTTP proxy → qBittorrent WebUI w/ tracker URL interception (DownloadHandler :767)	implemented	yes	integration test_bridge_header_rewrite.py	tested green suite
Theme injection into proxied	implemented	yes	integration test_bridge_theme_injection.py	tested green suite

Feature	Impl	Wired	Tests	Valid
WebUI (plugins/ theme_injector.py)				

2. qBitTorrent-go backend (Go/Gin) — opt-in --profile go

Entry: qBitTorrent-go/cmd/qbittorrent-proxy/main.go (routes registered :54-101). **Skeleton/rewrite-in-progress** (per AGENTS.md + docs/migration/PARITY_GAPS.md). NOT running by default.

2a. HTTP handlers (internal/api/*.go)

Feature	Impl	Wired	Tests	Valid
GET /health — HealthHandler (api/ health.go:5)	implemented	opt-in	go internal/api/ api_test.go	unit- (go t
GET /api/v1/config — ConfigHandler (api/ download.go:210)	implemented	opt-in	go coverage_test.go	unit-
GET /api/v1/bridge/ health — BridgeHealthHandler (api/download.go:189)	implemented	opt-in	go coverage_test.go	unit-
POST /api/v1/search — SearchHandler (api/ search.go:14)	implemented (proxies qBit search; no plugin fan- out)	opt-in	go api_test.go, coverage_test.go	unit-

Feature	Impl	Wired	Tests	Valid
POST /api/v1/search/ sync — SearchSyncHandler (api/search.go:50)	implemented	opt-in	go coverage_test.go	unit-
GET /api/v1/search/ stream/:id — SearchStreamHandler SSE (api/ search.go:91)	implemented	opt-in	go coverage_test.go	unit-
GET /api/v1/search/:id — GetSearchHandler (api/search.go:144)	implemented	opt-in	go getsearch_results_test.go, coverage_test.go	unit-
POST /api/v1/ search/:id/abort — AbortSearchHandler (api/search.go:177)	implemented	opt-in	go coverage_test.go	unit-
POST /api/v1/download — DownloadHandler (api/download.go:17)	partial — returns mock success, no qBit integration	opt-in	go coverage_test.go	unit- func hole
POST /api/v1/download/ file — DownloadFileHandler (api/download.go:41)	implemented	opt-in	go coverage_test.go	unit-
POST /api/v1/magnet — MagnetHandler (api/ download.go:86)	implemented	opt-in	go api_test.go, magnet_stress_chaos_test.go	unit stres chaos
GET /api/v1/downloads/ active — ActiveDownloadsHandler (api/download.go:147)	stub — returns empty	opt-in	go coverage_test.go	unit- func hole

Feature	Impl	Wired	Tests	Valid
	array, no qBit query			
POST /api/v1/auth/ qbittorrent — QBittorrentAuthHandler (api/download.go:156)	implemented	opt-in	go coverage_test.go	unit-
GET /api/v1/theme — GetThemeHandler (api/ theme.go:90)	implemented	opt-in	go theme_hardening_test.go, coverage_test.go	unit-
PUT /api/v1/theme — PutThemeHandler (api/ theme.go:96)	implemented	opt-in	go theme_hardening_test.go	unit-
GET /api/v1/hooks — ListHooksHandler (api/ hooks.go:91)	implemented (store only, no dispatch)	opt-in	go api_test.go, coverage_test.go	unit-
POST /api/v1/hooks — CreateHookHandler (api/hooks.go:97)	implemented (store only)	opt-in	go api_test.go	unit-
DELETE /api/v1/ hooks/:id — DeleteHookHandler (api/hooks.go:109)	implemented (store only)	opt-in	go api_test.go	unit-
GET /api/v1/schedules — ListSchedulesHandler (api/ scheduler_api.go:86)	implemented (store only)	opt-in	go coverage_test.go	unit- no d loop (RW sche neve

Feature	Impl	Wired	Tests	Validation	Video
POST /api/v1/schedules — CreateScheduleHandler (api/ scheduler_api.go:92)	implemented (store only)	opt-in	go coverage_test.go		unit- neve (RW-
DELETE /api/v1/ schedules/:id — DeleteScheduleHandler (api/ scheduler_api.go:104)	implemented (store only)	opt-in	go coverage_test.go		unit-

2b. qBittorrent Web API client (internal/client/*.go)

Feature	Impl	Wired	Tests	Validation	Video
NewClient — construct + login (client/ client.go:18)	implemented	opt-in	go client/ client_test.go	unit-only	N/A (no UI — go-test- covered client method; opt-in profile)
Login — POST / api/v2/auth/ login, store SID (client/ auth.go:11)	implemented	opt-in	go client/ auth_test.go	unit-only	N/A (no UI — go-test- covered client method; opt-in profile)
IsAuthenticated — SID presence (client/ auth.go:43)	implemented	opt-in	go client/ auth_test.go	unit-only	N/A (no UI — go-test- covered client method; opt-in profile)
GetSID — thread-safe SID accessor	implemented	opt-in	go client/ client_test.go	unit-only	N/A (no UI — go-test- covered

Feature	Impl	Wired	Tests	Validation	Video
(client/ client.go:39)					client method; opt-in profile)
GetTorrents — GET /api/v2/ torrents/info (client/ torrents.go:13)	implemented	opt-in	go client/ coverage_test.go	unit-only	N/A (no UI — go-test- covered client method; opt-in profile)
AddTorrent — POST /api/v2/ torrents/add URL (client/ torrents.go:30)	implemented	opt-in	go client/ coverage_test.go	unit-only	N/A (no UI — go-test- covered client method; opt-in profile)
AddTorrentFile — POST multipart /api/ v2/torrents/add (client/ torrents.go:54)	implemented	opt-in	go client/ coverage_test.go	unit-only	N/A (no UI — go-test- covered client method; opt-in profile)
GetAppVersion — GET /api/v2/ app/version (client/ torrents.go:87)	implemented	opt-in	go client/ coverage_test.go	unit-only	N/A (no UI — go-test- covered client method; opt-in profile)
StartSearch — POST /api/v2/ search/start (client/ search.go:27)	implemented	opt-in	go client/ coverage_test.go	unit-only	N/A (no UI — go-test- covered client method; opt-in profile)
GetSearchResults — GET /api/v2/ search/results	implemented	opt-in	go client/ coverage_test.go	unit-only	N/A (no UI — go-test- covered

Feature	Impl	Wired	Tests	Validation	Video
(client/ search.go:55)					client method; opt-in profile)
StopSearch — POST /api/v2/ search/stop (client/ search.go:84)	implemented	opt-in	go client/ coverage_test.go	unit-only	N/A (no UI — go-test- covered client method; opt-in profile)
SearchStatus — GET /api/v2/ search/status (client/ search.go:100)	implemented	opt-in	go client/ coverage_test.go	unit-only	N/A (no UI — go-test- covered client method; opt-in profile)
ListPlugins — GET /api/v2/ search/plugins (client/ search.go:121)	implemented	opt-in	go client/ coverage_test.go	unit-only	N/A (no UI — go-test- covered client method; opt-in profile)

2c. Services (internal/service/*.go)

Feature	Impl	Wired	Tests
Merge search orchestrator RunSearch poll/dedup loop (service/ merge_search.go:183)	implemented (qBit search only; no python-style plugin fan-out)	opt-in	go service/service_test.go, runsearch_dedup_test.go
StartSearch / GetSearchStatus / AbortSearch lifecycle (service/ merge_search.go:101-143)	implemented	opt-in	go getsearchstatus_hardening_test.go
Dedup key (FileURL→name+size) (service/ merge_search.go:269)	implemented	opt-in	go runsearch_dedup_test.go

Feature	Impl	Wired	Tests
FetchTorrent tracker fetch (service/ merge_search.go:276)	stub — returns “not yet implemented”	opt-in	none
Stats counters (service/ merge_search.go:280)	implemented	opt-in	go coverage_test.go
SSE broker pub/sub (service/sse_broker.go)	implemented (defined but not wired — handlers SSE inline)	opt-in	none dedicated
Metadata enricher (Go)	missing (RW-11)	no	none
Scheduler driver loop (Go)	missing (RW-10) — CRUD only, no cron execution	no	none

2d. webui-bridge binary (Go) + cross-cutting

Feature	Impl	Wired	Tests	Validation	Video
webui- bridge binary (cmd/ webui- bridge/ main.go)	partial (health + root only)	opt-in	none dedicated	not- validated	N/A (no UI of its own - proxy binary; opt in profile)
CORS + logging middleware (internal/ middleware/)	implemented	opt-in	go middleware/ middleware_test.go	unit-only	N/A (no UI — go-test- covered middlewar opt-in profile)
Models / data types (internal/ models/)	implemented	opt-in	go models/ models_test.go	unit-only	N/A (no UI — go-test- covered data types, opt-in profile)
Env config loader	implemented	opt-in	go config/ config_test.go	unit-only	N/A (no UI — go-test- covered

Feature	Impl	Wired	Tests	Validation	Video
(internal/ config/)					config loader; opt in profile)
Log redactor (internal/ logging/ redactor.go)	implemented	opt-in	go logging/ redactor_test.go	unit-only	N/A (no UI — go-test- covered redactor; opt-in profile)

3. boba-jackett (Go) — :7189

Entry: qBittorrent-go/cmd/boba-jackett/main.go; router internal/jackettapi/router.go. Owns Jackett credentials + indexer overrides + autoconfig run history; encrypted SQLite at /config/boba.db.

3a. HTTP endpoints (internal/jackettapi/router.go + handlers)

Feature	Impl	Wired	Tests	Validation
GET /healthz (router.go:60, health.go)	implemented	yes	go jackettapi/ health_test.go	unit (go)
GET /openapi.json (router.go:64, openapi.go)	implemented	yes	go jackettapi/ openapi_test.go; contract tests/contract/ openapi_test.go	unit + contract (go)
GET /api/v1/ jackett/ credentials — list (router.go:67, credentials.go)	implemented	yes	go credentials_test.go; integration tests/ integration/ jackett_db_test.go	unit + integrati (go)
POST /api/v1/ jackett/ credentials — create/update (router.go:67, credentials.go)	implemented	yes	go credentials_test.go	unit + integrati (go)

Feature	Impl	Wired	Tests	Validation
DELETE /api/v1/jackett/credentials/{name} (router.go:77, credentials.go)	implemented	yes	go credentials_test.go	unit + integrati (go)
GET /api/v1/jackett/indexers — list configured (router.go:86, indexers.go)	implemented	yes	go jackettapi/indexers_test.go	unit (go)
POST/PATCH/DELETE /api/v1/jackett/indexers/{id} (router.go:93, indexers.go)	implemented	yes	go indexers_test.go	unit (go)
POST /api/v1/jackett/indexers/{id}/test — test config (indexers.go:239)	implemented	yes	go indexers_test.go	unit (go)
GET /api/v1/jackett/catalog — indexer catalog (router.go:112, catalog.go)	implemented	yes	go jackettapi/catalog_test.go; benchmark catalog_bench_test.go	unit + bench (g
POST /api/v1/jackett/catalog/refresh (router.go:119, catalog.go)	implemented	yes	go catalog_test.go	unit (go)
GET /api/v1/jackett/	implemented	yes	go jackettapi/runs_test.go; e2e tests/	unit + e2 (go)

Feature	Impl	Wired	Tests	Validation
autoconfig/runs — run history (router.go:137, runs.go)			e2e/ jackett_management_test.go	
GET /api/v1/jackett/ autoconfig/runs/{id} — run detail (router.go:144, runs.go)	implemented	yes	go runs_test.go	unit (go)
POST /api/v1/jackett/ autoconfig/run — trigger run (router.go:151, runs.go)	implemented	yes	go runs_test.go; e2e tests/e2e/ jackett_management_test.go	unit + e2e (go)
GET/POST /api/v1/jackett/overrides (router.go:160, overrides.go)	implemented	yes	go jackettapi/ overrides_test.go	unit (go)
DELETE /api/v1/jackett/overrides/{env} (router.go:170, overrides.go)	implemented	yes	go overrides_test.go	unit (go)
Admin auth middleware (admin/admin on mutating routes) (jackettapi/auth_middleware.go)	implemented	yes	go auth_middleware_test.go	unit (go)
Hardened CORS middleware (jackettapi/cors_middleware.go)	implemented	yes	go cors_middleware_test.go	unit (go)

3b. Autoconfig engine + persistence (internal/jackett/, internal/db/)

Feature	Impl	Wired	Tests	Validation	Vi
Jackett autoconfig engine	implemented	yes	go jackett/ autoconfig_test.go, autoconfig_bench_test.go	unit + bench (go)	N/ — cov en

Feature	Impl	Wired	Tests	Validation	View
(jackett/ autoconfig.go)					ex by jac au run
Fuzzy indexer matcher (jackett/ matcher.go)	implemented	yes	go jackett/ matcher_test.go	unit (go)	N/ — cov ma
Jackett HTTP client (jackett/ client.go)	implemented	yes	go jackett/ client_test.go	unit (go)	N/ — cov cli
Encrypted SQLite — AES-256- GCM crypto (db/crypto.go)	implemented	yes	go db/crypto_test.go; security tests/security/ credential_leak_test.go	unit + security (go)	N/ — + s cov cry
SQLite migrations (db/ migrate.go)	implemented	yes	go db/migrate_test.go	unit (go)	N/ — cov mi
Credential / indexer / runs repos (db/repos/ *.go)	implemented	yes	go db/repos/*_test.go	unit (go)	N/ — cov rep
.env bootstrap import (bootstrap/ *.go)	implemented	yes	go bootstrap/ bootstrap_test.go	unit (go)	N/ — cov bo
Master-key ensure (first- boot) (bootstrap/ *.go)	implemented	yes	go bootstrap/ bootstrap_test.go	unit (go)	N/ — cov fir key en
.env file parse/write (envfile/*.go)	implemented	yes	go envfile/ parse_test.go, write_test.go	unit (go)	N/ — cov en O)

4. Tracker plugins (plugins/*.py)

Plugin contract: class with `url`, `name`, `supported_categories`, `search()`, `download_torrent()` — the *target* contract, not a description of every engine today (see CLAUDE.md § Plugin System for the measured exceptions). Installed into the container by `install-plugin.sh`. Three counts, each answering a different question:

- **43 search-plugin engines** named by the `PLUGINS=()` array — the canonical managed roster.
- **43 distinct engine modules** on disk across `plugins/`, `plugins/community/` and `plugins/webui_compatible/`; measured 2026-08-21, this is exactly the curated array.
- **36 *.py files** sit directly in `plugins/` — the engines plus support modules (`nova2`, `novaprinter`, `socks`, `helpers`, `env_loader`, `theme_injector`, `download_proxy`), which are NOT plugins.

Corrected per BOB-149: the previous revision stated 44 and 27, said “3 support” while listing 7, and claimed “many curated names have NO matching file in this repo tree”. That last claim was FALSE — every one of the 43 curated names resolves to a file in this tree (verified by set comparison, control-needle proven §11.4.201(7)(b)). These counts are guarded by `scripts/pre_build/check_cm_plugin_count.sh`.

4a. Plugins present in `plugins/` AND named in the curated array

Plugin	Type	Impl	Wired (curated?)	Tests
eztv.py	public	implemented	yes (curated)	stress test_plugin_p
limetorrents.py	public	implemented	yes (curated)	stress test_plugin_p multi-word tests/un. test_plugin_multiwor stress/chaos tests/s test_multiword_query (\$11.4.85)
piratebay.py	public	implemented		stress test_plugin_p

Plugin	Type	Impl	Wired (curated?)	Tests
			yes (curated)	
solidtorrents.py	public	implemented	yes (curated)	stress test_plugin_p
torlock.py	public	implemented	yes (curated)	stress test_plugin_p
rutor.py	public (anonymous)	implemented	yes (curated)	stress test_plugin_p
nyaa.py	public	implemented	yes (curated)	stress test_plugin_p
anilibra.py	public	implemented	yes (curated)	stress test_plugin_p

Plugin	Type	Impl	Wired (curated?)	Tests
bitsearch.py	public	implemented	yes (curated)	stress test_plugin_p
gamestorrents.py	public	implemented	yes (curated)	stress test_plugin_p
kickass.py	public	implemented (degrades to honest empty)	yes (curated)	stress test_plugin_p
megapeer.py	public	implemented	yes (curated)	stress test_plugin_p

Plugin	Type	Impl	Wired (curated?)	Tests
tokyotoshokan.py	public	implemented	yes (curated)	stress test_plugin.py
torrentgalaxy.py	public	implemented	yes (curated)	stress test_plugin.py
torrentkitty.py	public	implemented	yes (curated)	stress test_plugin.py
yts.py	public	implemented	yes (curated)	stress test_plugin.py
jackett.py	aggregator	implemented	yes (curated)	covered via jackett a

Plugin	Type	Impl	Wired (curated?)	Tests
iptorrents.py	private (freeleech-only)	implemented	yes (curated)	integration tests/integration_test_iptorrents.py
rutracker.py	private (cookies / CAPTCHA)	implemented	yes (curated)	integration test_tracker_injection tests/unit_tests (\$1.1); stress test_plugin_parsers (\$11.4.85); ReDoS-deadlock challenges/scripts/rutracker_redos_regression
kinozal.py	private	implemented	yes (curated)	integration test_tracker_injection

Plugin	Type	Impl	Wired (curated?)	Tests
nmclub.py	private (cookies)	implemented	yes (curated)	integration test_tra

4b. Plugin support modules

Module	Type	Impl	Wired	Tests	Valida
nova2.py — nova3 search harness	support	implemented	yes	covered via search e2e	tested- green-i suite
novaprinter.py — result emitter (NDJSON- patched)	support	implemented	yes	covered via search e2e	tested- green-i suite
helpers.py — HTTP/retrieve helpers	support	implemented	yes	covered via search e2e	tested- green-i suite
socks.py — SOCKS proxy support	support	implemented	yes	covered indirectly	not- validat

Module	Type	Impl	Wired	Tests	Validation
env_loader.py — env/credential loader for plugins	support	implemented	yes	covered via tracker auth tests	tested- green-i suite
Per-plugin download_torrent() path (.torrent fetch)	feature	implemented	yes	integration test_buttons_api.py (download/file)	tested- green-i suite
Freeleech-only download policy (IPTorrents [free] tag)	policy	implemented	yes	integration test_iporrents.py	tested- green-i suite

4c. Discrepancy rows (honest, §11.4.6)

Plugin	Type	Impl	Wired (curated?)	Tests	Validation	Vi
torrentproject / torrentscsv	public (curated names)	named in install- plugin.sh PLUGINS array but NO matching plugins/ *.py file in tree	curated- name-only	none	not- implemented- in-repo-tree (install script expects them elsewhere)	N/ im in — na plu
23 other curated names absent from tree: academictorrents, ali213, audiobookbay, bitru, bt4g,	public (curated names)	named in curated array; NO plugins/	curated- name-only	none	not- implemented- in-repo-tree (name in install array; file absent)	N/ im in — na plu

Plugin	Type	Impl	Wired (curated?)	Tests	Validation	Vi
btsow, extratorrent, glotorrents, linuxtracker, one337x, pctorrent, pirateiro, rockbox, snowfl, therarbg, torrentdownload, torrentfunk, xfsusb, yihua, yourbittorrent (+ torrentproject, torrentscsv)			*.py in this tree			

NOTE (§11.4.6, CORRECTION vs Rev 2): the install-plugin.sh PLUGINS array contains **44 entries**, not 12. Of those, ~21 have a matching plugins/*.py file in this repo; the rest are curated names with no file present in the tree. Stated as fact, not asserted as working.

5. Angular 21 frontend dashboard (frontend/)

Standalone SPA, signals-based; built to download-proxy/src/ui/dist/frontend, served from :7187. Vitest unit tests; Playwright e2e under frontend/e2e.

5a. Dashboard component controls (components/dashboard/dashboard.component.ts)

Feature	Impl	Wired	Tests
Search box (query signal) + submit	implemented	yes	vitest dashboard.component.spec.ts; py integration test_dashboard_rendering.py, test_ui_comprehensive.py

Feature	Impl	Wired	Tests
Result grid — sortable (name/type/size/seeds/ leechers/quality/sources)	implemented	yes	vitest dashboard.component.spec.ts
Results tab	implemented	yes	py test_ui_comprehensive.py
Active Downloads tab + view (dashboard.component.ts:745 loadDownloads)	implemented	yes	py test_buttons_api.py
Trackers tab + per-tracker auth chips (expandable)	implemented	yes	py test_auth_state_ui.py
Schedules tab	implemented	yes	py test_dashboard_automation.py
Hooks tab	implemented	yes	py test_dashboard_automation.py
Per-row Magnet button → magnet dialog trigger	implemented	yes	vitest magnet- dialog.component.spec.ts; py test_magnet_dialog.py
Per-row add-to-qBittorrent (qBit) button	implemented	yes	py test_button_functions.py, test_buttons_api.py

Feature	Impl	Wired	Tests
Per-row Download (.torrent) button	implemented	yes	py test_buttons_api.py
Per-row processing spinner (isBusy())	implemented	yes	vitest dashboard.component.spec.ts
Bridge health indicator + retry button	implemented	yes	py test_bridge_root_liveness.py
Tracker-stat dialog trigger	implemented	yes	py contract test_tracker_stats_contract.

5b. Dialog + UI components

Feature	Impl	Wired	Tests	
Magnet dialog — readonly textarea + Copy/Open/Add/Close (magnet-dialog.component.ts)	implemented	yes	vitest magnet-dialog.component.spec.ts; py test_magnet_dialog.py	t g s
	implemented	yes		

Feature	Impl	Wired	Tests	Validation
qBit login dialog — user/pass/remember/Login/Cancel (qbit-login-dialog.component.ts)			vitest qbit-login-dialog.component.spec.ts; py test_login_actions.py, test_auth_state_ui.py	unit (vitest)
Theme picker — palette dropdown + light/dark toggle (theme-picker.component.ts)	implemented	yes	vitest theme-picker.component.spec.ts; py e2e test_theme_runtime.py, test_crossapp_theme.py	unit (vitest)
Confirm dialog — title/message/Confirm/Cancel/backdrop (confirm-dialog.component.ts)	implemented	yes	vitest confirm-dialog.component.spec.ts	unit (vitest)
Tracker-stat dialog — JSON view + Copy + status badge + notes table + Esc-close (tracker-stat-dialog.component.ts)	implemented	yes	vitest tracker-stat-dialog.component.spec.ts; py contract test_tracker_stats_contract.py	unit (vitest)
Toast container — info/success/warning/error auto-dismiss (toast-container.component.ts)	implemented	yes	vitest toast-container.component.spec.ts	unit (vitest)
Site footer (site-footer.component.ts)	implemented	yes	vitest	unit (vitest)

5c. Services (frontend/src/app/services/)

Feature	Impl	Wired	Tests	Validation
API service — HTTP wrapper (search/download/magnet/auth/schedules/hooks/downloads/stats/config), 15s timeout, runtime baseUrl (api.service.ts)	implemented	yes	vitest api.service.spec.ts	unit (vitest)
SSE service — EventSource live result streaming	implemented	yes	vitest sse.service.spec.ts; py integration	

Feature	Impl	Wired	Tests	Validation
+ bearer-token (sse.service.ts)			test_realtime_streaming.py, test_streaming_browser.py	tested- green-in suite
Theme service — palette/mode + localStorage + cross-app sync (theme.service.ts)	implemented	yes	vitest theme.service.spec.ts; py e2e test_theme_runtime.py, test_crossapp_theme.py	tested- green-in suite
Toast service — toast queue state (toast.service.ts)	implemented	yes	vitest toast.service.spec.ts	unit (vitest)
Dialog service — promise-based confirm state (dialog.service.ts)	implemented	yes	vitest dialog.service.spec.ts	unit (vitest)

5d. Jackett page (frontend/src/app/jackett/)

Feature	Impl	Wired	Tests
Credentials — list + Add/Edit/Delete (per- row, window.confirm) + banners (jackett/ credentials/ credentials.component.ts)	implemented	yes	vitest credentials.component.spec.ts
Credential edit dialog — name/username/ password/cookies/Save/ Cancel (credential-edit- dialog.component.ts)	implemented	yes	vitest credential-edit- dialog.component.spec.ts

Feature	Impl	Wired	Tests
Indexers — Configured/Catalog/History tab container (jackett/indexers/indexers.component.ts)	implemented	yes	vitest indexers.component.spec.ts
Indexers — Configured tab (list + per-row remove)	implemented	yes	vitest indexers.component.spec.ts
Indexers — Catalog tab (list + per-row add)	implemented	yes	vitest indexers.component.spec.ts
Indexers — History tab (autoconfig run history)	implemented	yes	vitest indexers.component.spec.ts
Indexer add dialog — credential dropdown + Save/Cancel + error banner (indexer-add-dialog.component.ts)	implemented	yes	vitest indexer-add-dialog.component.spec.ts
IPTorrents cookie-flow component (iptorrents-cookie-flow.component.ts)	implemented	yes	vitest (jackett specs)

5e. Frontend infra

Feature	Impl	Wired	Tests	Validation	Video
Frontend coverage thresholds (40% lines/branches/funcs/stmts, v8)	infra	yes	vitest.config.ts	enforced in vitest config	N/A (no UI — build/coverage infra)

6. BobaLink browser extension (extension/)

WXT + TypeScript Manifest-V3. Detects magnet/.torrent links and forwards to merge service :7187. Status detail: docs/browser_extension/Status.md. Built zips at extension/.output/bobalink-1.0.0-{chrome,firefox}.zip.

6a. User-facing surfaces

Feature	Impl	Wired	Tests	Validation	Video
Popup — per-row Send button (src/popup/, entrypoints/popup.ts)	implemented	yes	a11y tests/a11y/popup.a11y.test.ts	tested-green-in-suite	VIDE CONF 2026- boba-e popup. docs/q record boba-e verdic
Popup — Send-All button	implemented	yes	a11y popup.a11y.test.ts	tested-green-in-suite	VIDE CONF 2026- boba-e popup. docs/q record boba-e verdic
Popup — Refresh button + status indicator (online/offline/warning)	implemented	yes	a11y popup.a11y.test.ts	tested-green-in-suite	VIDE CONF 2026- boba-e popup. docs/q record boba-e verdic
Popup — torrent list (name + type + infohash prefix + sent badge)	implemented	yes	a11y popup.a11y.test.ts	tested-green-in-suite	VIDE CONF 2026- boba-e popup. docs/q record boba-e verdic
Popup — Open Options link	implemented	yes	a11y popup.a11y.test.ts	tested-green-in-suite	VIDE CONF 2026- boba-e popup. docs/q record boba-e verdic
	implemented	yes			

Feature	Impl	Wired	Tests	Validation	Video
Options page (src/options/ options.ts)			ally options.ally.test.ts; integration options- background.test.ts	tested- green-in- suite	VIDEO CONF 2026- boba-e option docs/q record boba-e verdic
Content script — link/ text highlight overlay (src/ content/ highlight.ts, entrypoints/ content.ts)	implemented	yes	integration content- background.test.ts; security content- xss.test.ts	tested- green-in- suite	VIDEO CONF 2026- boba-e detect docs/q record boba-e verdic
Background service worker — message hub (src/ background/ entrypoints/ background.ts)	implemented	yes	integration tests/ integration/*- background.test.ts	tested- green-in- suite	N/A (n covere exerci extens popup
Context- menu entries (right-click torrent actions)	implemented	yes	integration *- background.test.ts	tested- green-in- suite	N/A (n covere exerci extens popup
Keyboard shortcuts (scan-now / highlight- toggle / send- all)	implemented	yes	integration *- background.test.ts	tested- green-in- suite	N/A (n covere exerci extens popup
Action badge (detected count) + notifications	implemented	yes	integration *- background.test.ts	tested- green-in- suite	N/A (n covere exerci extens popup

6b. Scanners (src/scanner/)

Feature	Impl	Wired	Tests	Validation	Video
	implemented	yes	vitest unit		

Feature	Impl	Wired	Tests	Validation	Video
Base scanner — computeStableId() lifecycle (scanner/base.ts)				tested-green-in-suite	N/A UI - test coverage scanner.exe by external scanner detection
Link scanner — magnet/.torrent (scanner/link-scanner.ts)	implemented	yes	security scanner-hostile-input.test.ts; perf scanner.perf.test.ts	tested-green-in-suite	N/A UI - test coverage scanner.exe by external scanner detection
Text scanner — bare magnet text (scanner/text-scanner.ts)	implemented	yes	security scanner-hostile-input.test.ts	tested-green-in-suite	N/A UI - test coverage scanner.exe by external scanner detection
Scanner orchestrator — dedup + mutation-observer re-scan (scanner/orchestrator.ts)	implemented	yes	perf orchestrator-scaling.perf.test.ts	tested-green-in-suite	N/A UI - test coverage scanner.exe by external scanner detection
Site DB — per-site selector match rules (scanner/site-db.ts)	implemented	yes	vitest unit	tested-green-in-suite	N/A UI - test coverage scanner.exe by external

Feature	Impl	Wired	Tests	Validation	Video
					scan detect

6c. Parsers (src/parser/)

Feature	Impl	Wired	Tests	Validation	Video
Bencode parser (parser/bencode.ts)	implemented	yes	security bencode-torrentfile-hostile.test.ts; perf parsers.perf.test.ts	tested-green-in-suite	N/A (no UI — test-covered parser; exercised by extension scan-detect)
Magnet parser (parser/magnet.ts)	implemented	yes	security infohash-detection-hostile.test.ts; perf magnet.perf.test.ts	tested-green-in-suite	N/A (no UI — test-covered parser; exercised by extension scan-detect)
Torrent-file parser (parser/torrent-file.ts)	implemented	yes	security bencode-torrentfile-hostile.test.ts	tested-green-in-suite	N/A (no UI — test-covered parser; exercised by extension scan-detect)

6d. API client + offline (src/api/)

Feature	Impl	Wired	Tests	Validation	Video
Boba API client — POST /api/v1/download, retry + rate-limit (api/	implemented	yes	vitest unit; chaos tests/chaos/boba-client-resilience.chaos.test.ts	tested-green-in-suite	N/A (no UI — test-covered API client)

Feature	Impl	Wired	Tests	Validation	Video
boba-client.ts)					exe by exte Sen N/A UI - test cov hea pro exe by exte stat ind
Health probe — probeHealth() (api/health.ts)	implemented	yes	vitest unit	tested-green-in-suite	N/A UI - test cov offl que cha cov
Offline send queue — IndexedDB enqueue + re-dispatch (api/queue.ts)	implemented	yes	chaos tests/chaos/queue.chaos.test.ts	tested-green-in-suite	

6e. Shared utilities + tabgroups (src/shared/, src/tabgroups/)

Feature	Impl	Wired	Tests	Validation	Video
Crypto — AES-256-GCM token encrypt/decrypt (shared/crypto.ts)	implemented	yes	security crypto-tamper.test.ts, no-hardcoded-secret.test.ts	tested-green-in-suite	N/A (r UI — cover crypto exerci by exten Send token path)
Storage — namespaced chrome.storage.local wrapper (shared/storage.ts)	implemented	yes	vitest unit	tested-green-in-suite	N/A (r UI — cover storag wrapp
Constants — site selectors / rate-limit / retry config	implemented	yes	covered via scanner tests	tested-green-in-suite	N/A (r UI — cover consta

Feature	Impl	Wired	Tests	Validation	Video
(shared/ constants.ts)					
Errors — Network/ Server/Storage error types (shared/ errors.ts)	implemented	yes	vitest unit	tested- green-in- suite	N/A (no UI — coverage error types)
Events — TypedEventEmitter bus (shared/ events.ts)	implemented	yes	vitest unit	tested- green-in- suite	N/A (no UI — coverage event bus)
Logger — namespaced, no- secret (shared/ logger.ts)	implemented	yes	vitest unit	tested- green-in- suite	N/A (no UI — coverage logger)
Utils — debounce / TokenBucket rate limiter (shared/ utils.ts)	implemented	yes	vitest unit	tested- green-in- suite	N/A (no UI — coverage utils/r limite
Tab groups — batch grouped-tab torrents (tabgroups/ index.ts)	implemented	yes	vitest unit (tab-group Challenge per ext Status)	tested- green-in- suite	N/A (no UI of own — test- coverage tab-gr batch exerci by exten Send-

6f. i18n locales (extension/src/public/_locales/)

Feature	Impl	Wired	Tests	Validation	Video
Locale en (English)	implemented	yes	i18n tests/ i18n/locale- safety.test.ts	tested- green-in- suite	N/A (no UI of its own — i18n locale data; test- covered; surfaced inside extension

Feature	Impl	Wired	Tests	Validation	Video
Locale de (German)	implemented	yes	i18n locale- safety.test.ts	tested- green-in- suite	popup/ options) N/A (no UI of its own — i18n locale data; test- covered; surfaced inside extension popup/ options) N/A (no UI of its own — i18n locale data; test- covered; surfaced inside extension popup/ options)
Locale es (Spanish)	implemented	yes	i18n locale- safety.test.ts	tested- green-in- suite	N/A (no UI of its own — i18n locale data; test- covered; surfaced inside extension popup/ options)
Locale fr (French)	implemented	yes	i18n locale- safety.test.ts	tested- green-in- suite	N/A (no UI of its own — i18n locale data; test- covered; surfaced inside extension popup/ options)
Locale it (Italian)	implemented	yes	i18n locale- safety.test.ts	tested- green-in- suite	N/A (no UI of its own — i18n locale data; test- covered;

Feature	Impl	Wired	Tests	Validation	Video
Locale ja (Japanese)	implemented	yes	i18n locale- safety.test.ts	tested- green-in- suite	surfaced inside extension popup/ options) N/A (no UI of its own — i18n locale data; test- covered; surfaced inside extension popup/ options) N/A (no UI of its own — i18n locale data; test- covered; surfaced inside extension popup/ options)
Locale pt (Portuguese)	implemented	yes	i18n locale- safety.test.ts	tested- green-in- suite	N/A (no UI of its own — i18n locale data; test- covered; surfaced inside extension popup/ options) N/A (no UI of its own — i18n locale data; test- covered; surfaced inside extension popup/ options)
Locale ru (Russian)	implemented	yes	i18n locale- safety.test.ts	tested- green-in- suite	N/A (no UI of its own — i18n locale data; test- covered; surfaced inside extension popup/ options)

6g. Live round-trip

Feature	Impl	Wired	Tests	Validation
Live download endpoint round-trip (tests/live/download-endpoint.live.test.ts)	implemented	yes (live-gated)	live download-endpoint.live.test.ts	partial — operator-gated live env

7. WebUI bridge (webui-bridge.py) — host process :7188

Bridges qBittorrent WebUI with private-tracker auth. NOT a container.

Feature	Impl	Wired	Tests	Valid
WebUI proxy + header rewrite (webui-bridge.py)	implemented	host	integration test_webui_bridge.py, test_bridge_header_rewrite.py; stress test_bridge_stress_chaos.py	tested green suite
Private-tracker auth bridging (rutracker/kinozal/nmclub/iptorrents)	implemented	host	integration test_webui_bridge.py	partial live-g
Theme injection into bridged WebUI (plugins/theme_injector.py)	implemented	host	integration test_bridge_theme_injection.py	tested green suite
Bridge root liveness (/)	implemented	host	integration test_bridge_root_liveness.py	

Feature	Impl	Wired	Tests	Validation	Video
					tested green suite

8. Infrastructure / CLI / shell scripts

8a. boba-ctl CLI orchestrator (Go: cmd/boba-ctl/main.go + wrapper scripts/boba-ctl.sh)

Feature	Impl	Wired	Tests	Validation	Video
boba-ctl up — start stack (cmd/boba-ctl/main.go:32)	implemented	host	go cmd/boba-ctl/projectroot_test.go	tested-green-in-suite	N/A (Covered by boba-ctl demo, orchestrator demo.mp4)
boba-ctl down — stop stack (main.go:34)	implemented	host	go cmd/boba-ctl/projectroot_test.go	tested-green-in-suite	N/A (Covered by boba-ctl demo, orchestrator demo.mp4)
boba-ctl status — service status (main.go:36)	implemented	host	go cmd/boba-ctl/projectroot_test.go	tested-green-in-suite + LIVE	VIDEO CONF 2026-01-01 boba-ctl orchestrator demo.mp4 docs/qa/recordings/boba-ctl
boba-ctl health — health probe (main.go:38, cmd/boba-ctl/health_test.go)	implemented	host	go cmd/boba-ctl/health_test.go	tested-green-in-suite + LIVE	VIDEO CONF 2026-01-01 boba-ctl orchestrator demo.mp4
boba-ctl list — list services (main.go:40)	implemented	host	go cmd/boba-ctl/projectroot_test.go	tested-green-in-suite + LIVE	VIDEO CONF 2026-01-01 boba-ctl

Feature	Impl	Wired	Tests	Validation	Video
boba-ctl.sh wrapper — compose- compat shim (up/down/ps/ config/pull + passthrough) (scripts/boba- ctl.sh)	implemented	host	covered via boba- ctl go tests	tested- green-in- suite	orchest demo.mp N/A (C test-co exercis ctl CLI

8b. Lifecycle scripts (repo root)

Feature	Impl	Wired	Tests	Valid
start.sh — container bring-up + Jackett key extraction/ injection	implemented	host	covered via e2e test_fixtures_bring_up_services.py	tested green suite
stop.sh — stop/remove/ purge	implemented	host	none dedicated	not- valida
setup.sh — one-time setup	implemented	host	none dedicated	not- valida
start-proxy.sh — proxy container entry helper	implemented	host	none dedicated	not- valida
install- plugin.sh — copy curated plugins → container engines	implemented	host	covered indirectly (e2e search)	tested green suite
init-qbit- password.sh — qBit admin password bootstrap	implemented	host	none dedicated	not- valida

Feature	Impl	Wired	Tests	Valid
fix-qbit-password.sh — qBit admin password repair	implemented	host	none dedicated	not-validated
setup-webui-bridge-service.sh + webui-bridge.service — systemd unit for bridge	implemented	host	none dedicated	not-validated

8c. Test / CI wrappers (repo root)

Feature	Impl	Wired	Tests	Validation
ci.sh — manual CI (syntax+unit+integration+e2e+health)	implemented	host	self (is the gate)	tested-green-in-suite
run-all-tests.sh — full suite (hardcoded podman)	implemented	host	self	partial — fails on docker-only host (known gotcha)
test.sh — quick validation wrapper	implemented	host	self	tested-green-in-suite
test-all.sh — full validation wrapper	implemented	host	self	tested-green-in-suite

Feature	Impl	Wired	Tests	Validation
test-full.sh — extended validation wrapper	implemented	host	self	tested-green-in-suite

8d. scripts/ helpers

Feature	Impl	Wired	Tests
scripts/add-submodules.sh — submodule bootstrap	implemented	host	none dedicated
scripts/audit-plugins.sh — plugin discrepancy audit	implemented	host	none dedicated
scripts/build-releases.sh — build extension/Go release artifacts	implemented	host	none dedicated
scripts/codegraph_validate.sh — CodeGraph index validation (§11.4.78)	implemented	host	self
scripts/workable-items-export.sh — docs regeneration wrapper (§11.4.93/§11.4.65; renamed 2026-08-15 BOB-104 from docs_chain.sh — that name was a misnomer)	implemented	host	self
constitution/submodules/docs_chain/docs_chain — real §11.4.106 Docs Chain engine (Go binary, pinned to helixcode-v1.1.0)	implemented	host	self

Feature	Impl	Wired	Tests
scripts/ensure-macos-tunnel.sh — 0.0.0.0 SSH tunnel	implemented	host	none dedicated (log qa-resu tunnel-keepalive.log)
scripts/tunnel-keepalive.sh — tunnel self-heal	implemented	host	none dedicated
scripts/freeze-openapi.sh — freeze boba-jackett OpenAPI snapshot	implemented	host	contract tests/contract/openapi_test.go
scripts/generate_markdown_exports.sh — md→html/pdf export (§11.4.65)	implemented	host	self
scripts/helixqa.sh — HelixQA harness driver	implemented	host	self
scripts/opencode-helixqa.sh — OpenCode HelixQA driver	implemented	host	self
scripts/install_git_hooks.sh — git hook installer (§11.4.75)	implemented	host	self
scripts/pre_build_verification.sh — pre-build gate sweep	implemented	host	self (is the gate)
scripts/pre_code_review.sh — code-review gate (§11.4.125)	implemented	host	self
	implemented	host	self

Feature	Impl	Wired	Tests
scripts/ run_all_challenges.sh — challenge bank runner			
scripts/run-tests.sh — test runner helper	implemented	host	self
scripts/scan.sh — repo scan helper	implemented	host	none dedicated
scripts/track-build- resources.sh — build resource sampler (§11.4.24)	implemented	host	none dedicated
scripts/load-tracker- cookies.sh — per-tracker Netscape cookies-file autoload → .env (operator mandate 2026-08-15, 619a5f6)	implemented	host	challenge challenges/script credentials_wired_challenge
scripts/boba-svc.sh — sudo- free systemd-linger service wrapper (up/down/status/ logs)	implemented	host	challenge challenges/script install_sh_idempotent_challe credentials_wired_challenge
scripts/commit-push-all.sh — §11.4.234 dedicated commit/push entrypoint (hook-validation stage + always-unblocked push, 08e870e)	implemented	host	none dedicated

8e. Infra / compose / startup

Feature	Impl	Wired	Tests	Valida
docker- compose.yml — 2-container default + Go profile + boba- jackett service	infra	yes	e2e test_live_containers.py	tested- green- suite

Feature	Impl	Wired	Tests	Valida
Jackett auto-configuration at startup (key extract → inject JACKETT_API_KEY)	implemented	yes	integration test_jackett_autoconfig_real.py	tested- green- suite
BOBA_MASTER_KEY auto-generation (first-boot) for encrypted boba.db	implemented	yes	go bootstrap/bootstrap_test.go	unit (g

Components NOT inventoried / discrepancies (honest gaps, §11.4.6)

- **Rev 9 scope boundary (BOB-083, 2026-08-20):** this pass corrected 2 pre-existing rows (RW-06 rutracker ReDoS-deploy row §4b, RW-07 nnmclub-status row §1b) to PASS – LIVE with real evidence citations (1c0389a/BOB-093, 7baef2b/BOB-092), but did **NOT** re-audit the other ~290 pre-existing rows against the 161-commit delta since Rev 8 (2026-08-18) — same bounded-scope reasoning as Rev 8. Known-real, non-feature work that landed in that window and is intentionally NOT itemized as new rows here (process-safety/CI discipline, not a product feature): the BOB-126 killpg(-1, SIGKILL) forced-logout incident chain root-cause + fix (ad4b46a, new universal §11.4.263 anchor) and its surrounding gate/test hardening (BOB-127..135, ~150 commits). A future pass MAY re-run the full inventory to close this gap properly; until then this document is accurate for what it explicitly claims and honestly incomplete for what it does not.
- **Rev 8 scope boundary (BOB-075, 2026-08-18):** this pass added 3 new rows for scripts that shipped with no prior row (§8d) and fixed the missing Rev/date bump for the ea86ce1 content edit, but did **NOT** re-audit the other ~289 pre-existing rows against the current tree — that would be a full re-run of the original READ-ONLY repo-inventory process (§11.4.118 discovery-pressure), out of scope for a bounded staleness fix. Known-real capabilities that shipped since Rev 7 and are **not yet itemized at row-granularity** here (tracked as an honest gap, not silently assumed covered): the systemd service topology (boba.target, boba-stack.service, boba-webui-bridge.service, 444d94e/bbc63ca), outbound egress-proxy routing for tracker fetches (BOBA_UPSTREAM_PROXY, b44a2e4/4e62eca/be5062d), and the submodules/jackett version bump to v0.24.2406 (99a486e, BOB-076 — a dependency-version change, not a new

feature row). A future pass MAY re-run the full inventory; until then this document is accurate for what it explicitly claims and honestly incomplete for what it does not.

- **CORRECTION vs Rev 2:** boba-ctl.sh DOES exist at scripts/boba-ctl.sh (Rev 2 said it did not). Its subcommands are now itemized in §8a. The underlying Go binary is cmd/boba-ctl/main.go (5 subcommands: up/down/status/health/list).
- **CORRECTION vs Rev 2:** the install-plugin.sh PLUGINS curated array has **44 entries**, not 12. ~21 have a matching plugins/*.py file; the rest (incl. torrentproject, torrentscsv) are curated names with no file present in this tree — itemized in §4c, not asserted as working.
- **Prometheus/OpenTelemetry metrics** — tests/observability/test_metrics_exist.py + observability/ dir exist, but no prometheus_client Counter()/Histogram()/Gauge() definitions were found in download-proxy/src; metrics surface is via GET /api/v1/stats (counters maintained in-process), not a /metrics Prometheus endpoint. Stated as fact, not assumed.
- **Go DownloadHandler / ActiveDownloadsHandler** — flagged partial/stub in §2a (mock success / empty array, no qBit integration) per source reading. The Go profile is opt-in and NOT the shipped product.
- **Go scheduler** — CRUD-only, no driver loop (RW-10); schedules never fire. The Go enricher is missing (RW-11).
- **Video-recording confirmation** — PENDING for all rows EXCEPT the boba-ctl CLI subcommands (status/health/list) and the web-UI flows already filmed (POST /api/v1/search journey + dashboard tab/button/theme/Jackett tour). Remaining per-feature flows are an in-progress recording pass (§11.4.107/§11.4.143).